Q5: COUNT THE WORK DONE BY A LOOP

CODE:

```
#include <stdio.h>

unsigned long checks = 0;

int is_prime(unsigned long n)
{
   unsigned long i;

   checks = 0;

   if (n < 2)
      return 0;

   for (i = 2; i <= n / i; i++)
   {
      checks++;

      if (n % i == 0)
         return 0;
   }

   return 1;
}

int is_prime_while(unsigned long n)
```

```
{
   unsigned long i = 2;

   checks = 0;

   if (n < 2)
      return 0;

   while (i <= n / i)
   {
      checks++;

      if (n % i == 0)
         return 0;

      i++;
   }

   return 1;
}

int is_prime_do(unsigned long n)
{
   unsigned long i = 2;

   checks = 0;

   if (n < 2)
```

```
    return 0;

  do
  {
    if (i > n / i)
      break;

    checks++;

    if (n % i == 0)
      return 0;

    i++;

  } while (1);

  return 1;
}

int main()
{
  unsigned long tickets[] = {97, 100, 7919, 1000003};
  int i, p;

  printf("FOR LOOP:\n");

  for (i = 0; i < 4; i++)
  {
```

```
    p = is_prime(tickets[i]);

    printf("%lu %s divisions = %lu\n",
        tickets[i],
        p ? "prime" : "not prime",
        checks);
}

printf("\nWHILE LOOP:\n");

for (i = 0; i < 4; i++)
{
    p = is_prime_while(tickets[i]);

    printf("%lu %s divisions = %lu\n",
        tickets[i],
        p ? "prime" : "not prime",
        checks);
}

printf("\nDO-WHILE LOOP:\n");

for (i = 0; i < 4; i++)
{
    p = is_prime_do(tickets[i]);

    printf("%lu %s divisions = %lu\n",
        tickets[i],
```

```
            p ? "prime" : "not prime",
            checks);
    }

    return 0;
}
```

